

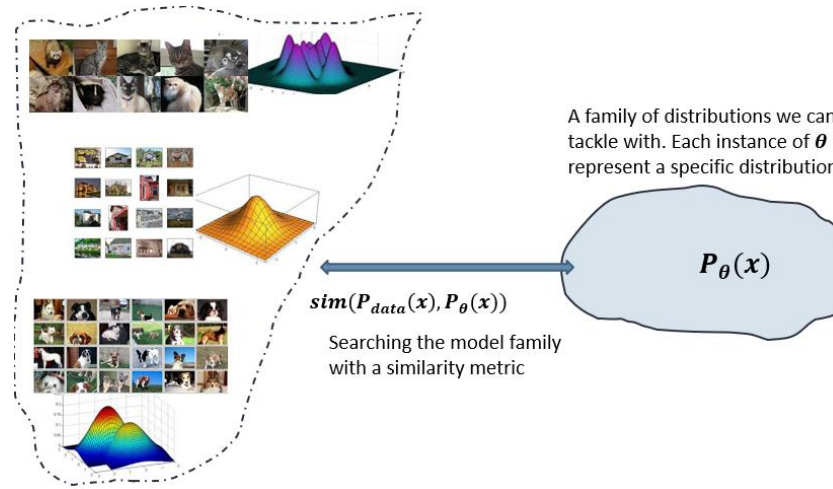


Diffusion Models

22-808: Generative models
Sharif University of Technology
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Fatemeh Seyyedsalehi

Recap



- ▶ We need a framework to interact with distributions for statistical generative models.
 - ▶ Probabilistic graphical models
 - ▶ Deep generative models
 - ▶ Autoregressive models
 - ▶ Variational Autoencoders
 - ▶ Generative adversarial networks
 - ▶ Normalizing Flow
 - ▶ Energy-based models
 - ▶ Score-based models
 - ▶ **Diffusion models**

Finally the last one 😊 what is the next ?

DDPM paper

- ▶ The main idea behind DALLÉ-2, stable diffusion 😊

Denoising Diffusion Probabilistic Models

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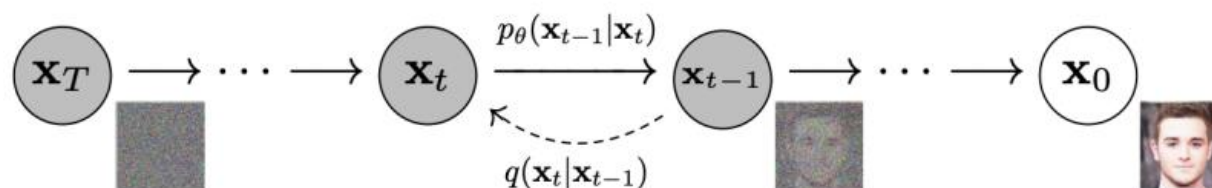
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Abstract

We present high quality image synthesis results using diffusion probabilistic models, a class of latent variable models inspired by considerations from nonequilibrium thermodynamics. Our best results are obtained by training on a weighted variational bound designed according to a novel connection between diffusion probabilistic models and denoising score matching with Langevin dynamics, and our models naturally admit a progressive lossy decompression scheme that can be interpreted as a generalization of autoregressive decoding. On the unconditional CIFAR10 dataset, we obtain an Inception score of 9.46 and a state-of-the-art FID score of 3.17. On 256x256 LSUN, we obtain sample quality similar to ProgressiveGAN. Our implementation is available at <https://github.com/hojonathanho/diffusion>.

Diffusion models



Ho, Jain, and Abbeel, Neurips 2020

Want to model the distribution of $\mathbf{x} \sim p$, where $\mathbf{x} \in \mathbb{R}^d$.

Construct a Markov chain $\mathbf{x}_0, \dots, \mathbf{x}_T \in \mathbb{R}^d$.

Learn a transition model, e.g. $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}|\mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))$.

Base case: $p(\mathbf{x}_T) = \mathcal{N}(0, I)$.

Marginal distribution over $\mathbf{x}_0$ is $p_\theta(\mathbf{x}_0) = \int p_\theta(\mathbf{x}_0, \dots, \mathbf{x}_T) d\mathbf{x}_1 \dots d\mathbf{x}_T$.

Learn the parameters so that $p(\mathbf{x}_0) \approx p_\theta(\mathbf{x}_0)$.

Diffusion models

Maximize the likelihood?

Want to model the distribution of $\mathbf{x} \sim p$, where $\mathbf{x} \in \mathbb{R}^d$.

Marginal distribution over $\mathbf{x}_0$ is $p_\theta(\mathbf{x}_0) = \int p_\theta(\mathbf{x}_0, \dots, \mathbf{x}_T) d\mathbf{x}_1 \dots d\mathbf{x}_T$.

Learn the parameters so that $p(\mathbf{x}_0) \approx p_\theta(\mathbf{x}_0)$.

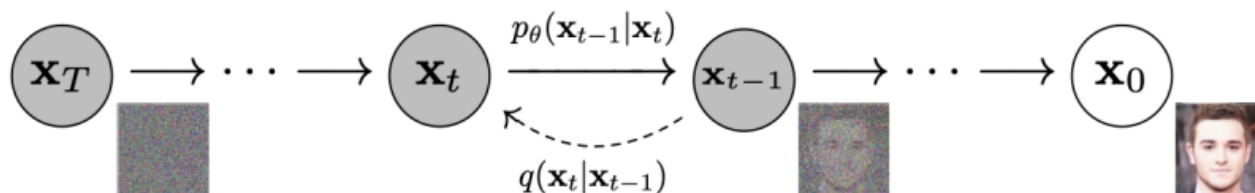
Likelihood is intractable:

$$\arg \max_{\theta} \mathbb{E}_{\mathbf{x}_0 \sim p} [\log p_\theta(\mathbf{x}_0)] = \mathbb{E}_{\mathbf{x}_0 \sim p} \left[\log \int p_\theta(\mathbf{x}_0, \dots, \mathbf{x}_T) d\mathbf{x}_1 \dots d\mathbf{x}_T \right].$$

Use the variational approximation!

Diffusion models

Using ELBO



Want to model the distribution of $\mathbf{x} \sim p$, where $\mathbf{x} \in \mathbb{R}^d$.

Marginal distribution over $\mathbf{x}_0$ is $p_\theta(\mathbf{x}_0) = \int p_\theta(\mathbf{x}_0, \dots, \mathbf{x}_T) d\mathbf{x}_1 \dots d\mathbf{x}_T$.

Fix $q_t(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t I)$ (hyper-parameters β_t).

The evidence lower-bound (Jensen):

$$\arg \max_{\theta} \mathbb{E}_{\mathbf{x}_0 \sim p} [\log p_\theta(\mathbf{x}_0)] \geq \mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \mathbf{x}_{1:T} \sim q}} \left[\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right].$$

Diffusion models

Using ELBO

The forward process: $q_t(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t I)$.

The reverse process: $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}|\mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))$.

$$\begin{aligned}\mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \mathbf{x}_{1:T} \sim q}} \left[\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] &= \mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \mathbf{x}_{1:T} \sim q}} \left[-\log p(\mathbf{x}_T) - \sum_{t>0} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)} \right] \\ &= \mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \mathbf{x}_{1:T} \sim q}} \left[-\log p(\mathbf{x}_T) - \sum_{t>0} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \right] \\ &= \mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \mathbf{x}_{1:T} \sim q}} \left[-\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} - \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right]\end{aligned}$$

Acti

Diffusion models

Using ELBO

The forward process: $q_t(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t I)$.

The reverse process: $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}|\mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))$.

Conditionals have closed form: $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t I)$

Where: $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\tilde{\alpha}_t}\beta_t}{1 - \tilde{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1 - \tilde{\alpha}_{t-1})}{1 - \tilde{\alpha}_t}\mathbf{x}_t$.

And: $\alpha_t = 1 - \beta_t$, $\tilde{\alpha}_t = \prod_{s=1}^t \alpha_s$, $\tilde{\beta}_t = \frac{1 - \tilde{\alpha}_{t-1}}{1 - \tilde{\alpha}_t}\beta_t$.

Diffusion models

Unpack the ELBO

The reverse process: $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}|\mu_{\theta}(\mathbf{x}_t, t), \Sigma_{\theta}(\mathbf{x}_t, t))$.

The ELBO: $\mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \mathbf{x}_{1:T} \sim q}} \left[-\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} - \sum_{t>1} \log \frac{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} - \log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1) \right]$

Optimization of the parameters decomposes term-wise:

$$\begin{aligned} \mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \mathbf{x}_{1:T} \sim q}} \left[-\sum_{t>1} \log \frac{p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] &= \mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \mathbf{x}_t \sim q_t}} [D(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))] \\ &= \mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \mathbf{x}_t \sim q_t}} \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_{\theta}(\mathbf{x}_t, t)\|^2 \right] + C. \end{aligned}$$

Acti

Reparameterization

We can directly compute $q_t(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\tilde{\alpha}_t}\mathbf{x}_0, (1 - \tilde{\alpha}_t)I)$.

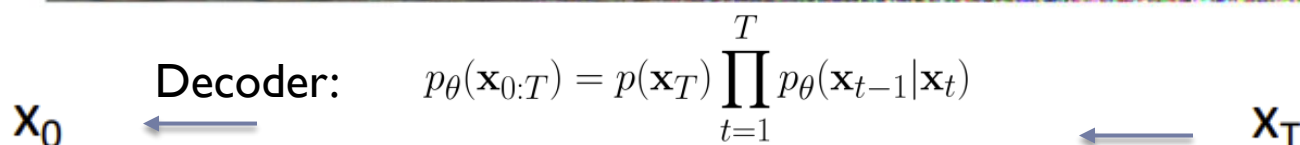
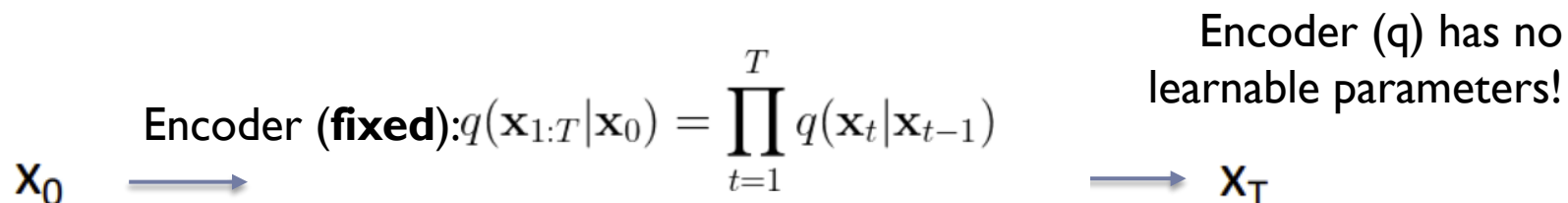
Define $x_t(\mathbf{x}_0, \varepsilon) = \sqrt{\tilde{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \tilde{\alpha}_t}\varepsilon$.

Re-parameterize $\mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta}{\sqrt{1 - \tilde{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right)$.

$$\begin{aligned} \mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \mathbf{x}_{1:T} \sim q}} \left[- \sum_{t>1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] &= \mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \varepsilon \sim \mathcal{N}(0, I)}} \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|^2 \right] + C \\ &= \mathbb{E}_{\substack{\mathbf{x}_0 \sim p \\ \varepsilon \sim \mathcal{N}(0, I)}} \left[\frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \tilde{\alpha}_t)} \|\varepsilon - \varepsilon_\theta(x_t(\mathbf{x}_0, \varepsilon), t)\|^2 \right] + C. \end{aligned}$$

Activ

Denoising diffusion probabilistic model



ELBO loss (negative ELBO averaged over data distribution):

$$\mathbb{E}_{q(\mathbf{x}_0)} [-\log p_{\theta}(\mathbf{x}_0)] \leq \mathbb{E}_{q(\mathbf{x}_0)q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[-\log \frac{p_{\theta}(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] =: L$$

Training and inference

Algorithm 1 Training

```
1: repeat
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ 
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$ 
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
5:   Take gradient descent step on
        $\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2$ 
6: until converged
```

Denoising score
matching training

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
```

Iteratively Sample from decoders $p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)$
(Annealed Langevin Dynamics)